

Multiple Choice

1. **Answer:** D

Options: A) First two digits from right to left; B) Median of all digits; C) First digit from right to left; D) First digit from left to right; E) Mode of all digits

Note: Correct option: D - First digit from left to right

2. **Answer:** A

Options: A) 100 N.; B) 0.01 N.; C) 10 N.; D) 0.5 N.; E) 0.001 N.

Note: Correct option: A - 100 N.

3. **Answer:** A

Options: A) 55; B) 50; C) 75; D) 85; E) 65

Note: Correct option: A - 55

4. **Answer:** A

Options: A) $v(0^+) = -4, (dv/dt)_-(0)^+ = 2.25$; B) $v(0^+) = -2, (dv/dt)_-(0)^+ = 3.25$; C) $v(0^+) = -3, (dv/dt)_-(0)^+ = 4.25$; D) $v(0^+) = 0, (dv/dt)_-(0)^+ = 1.5$; E) $v(0^+) = -3, (dv/dt)_-(0)^+ = 1.25$

Note: Correct option : A - $v(0^+) = -4, (dv/dt)_-(0)^+ = 2.25$

4. **Answer:** B

Options: A) $f(t) = [e^{-2t} + 2.24e^{-2t}\sin(5t - 26.6^\circ)]u(t)$; B) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$; C) $f(t) = [e^{-2t} - 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$; D) $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t + 26.6^\circ)]u(t)$

Note: Correct option : B - $f(t) = [e^{-2t} + 2.24e^{-2t}\cos(5t - 26.6^\circ)]u(t)$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: $-48e^{-3t} - (68/2)e^{-2t} + (33/2)e^{-4t}$.

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 82.2 in. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 5/36. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key